

Signal: antenna -> air-gap RC filter -> electrometer (G = 101) -> anti-alias + LMP7715 driver

-> PCM1804 (24 bit, 192 kHz) -> CS8406 -> S22083 -> AES3 110 R on RJ45 (STP cable, 100 m).

Supply: 6.98 V from the two-bucket PSU (or a 9-15 V battery) -> ADM7150 +5VA and +3V3.

Board: 200 x 100 mm, 4 layers, three compartments under an aluminium shield (layout.py).

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CERN-OHL-W-2.0

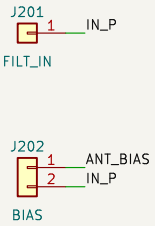
ELARA -- ELF Atmospheric Radio Analyser

Sheet: /
File: antenna_amplifier.kicad_sch

Title: ELARA antenna amplifier

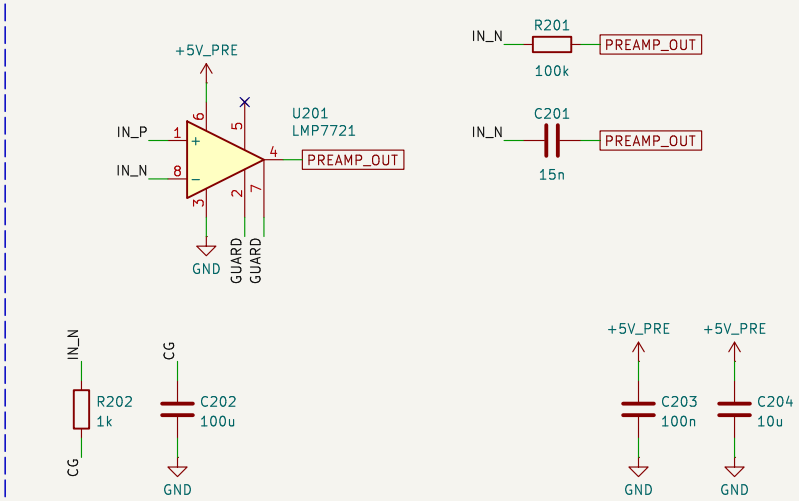
Size: A4	Date: 2026-09-30	**Rev: 0.2**
KiCad E.D.A. 9.0.9		Id: 1/5

INPUT (comp. 1)

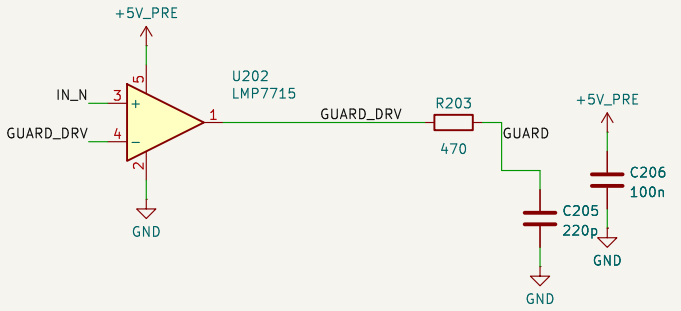


IN_P: guard on all layers,
no mask, PTFE turret.
J202: fit the G-ohm bias R.

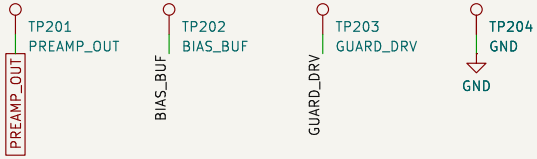
ELECTROMETER G = 101



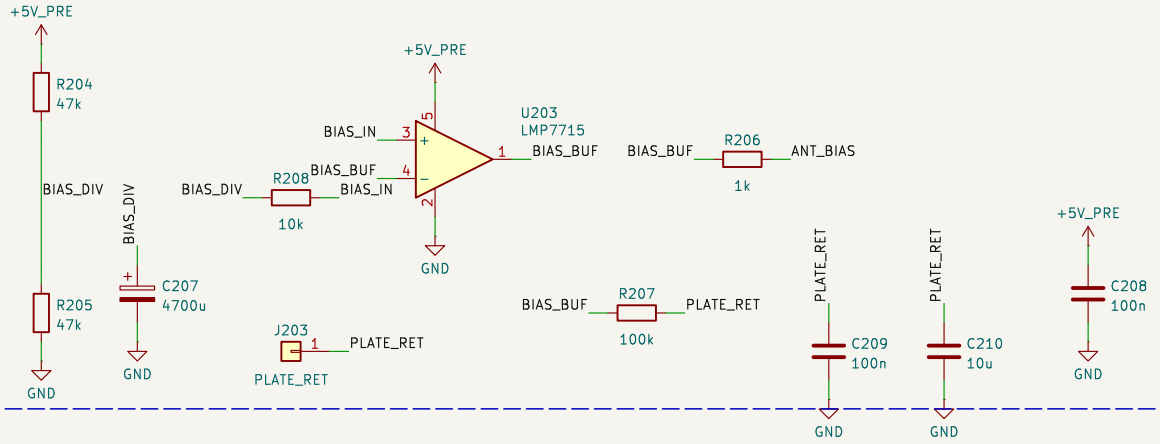
GUARD DRIVER



TEST POINTS



ANTENNA BIAS 2.5 V



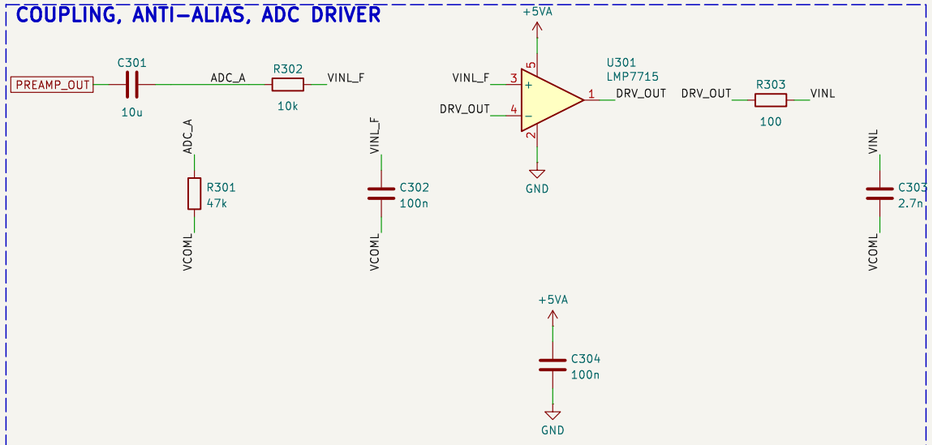
Cg returns to GND: the 2.5 V buffer noise stays out of the gain path.
Guard buffer senses IN− (virtual short to IN+): no bias current on IN_P.

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CERN-OHL-W-2.0
ELARA -- ELF Atmospheric Radio Analyser
Sheet: /Electrometer front end/
File: frontend.kicad_sch

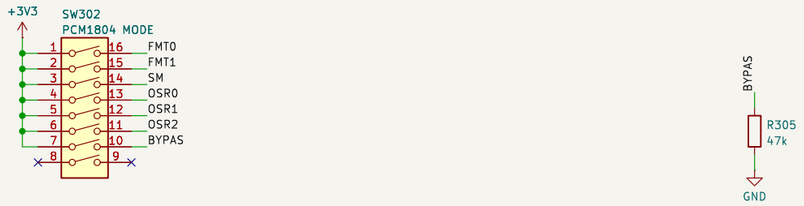
Title: ELARA antenna amplifier — Electrometer front end

Size: A4	Date: 2026-09-30	**Rev: 0.2**
KiCad E.D.A. 9.0.9		Id: 2/5

COUPLING, ANTI-ALIAS, ADC DRIVER

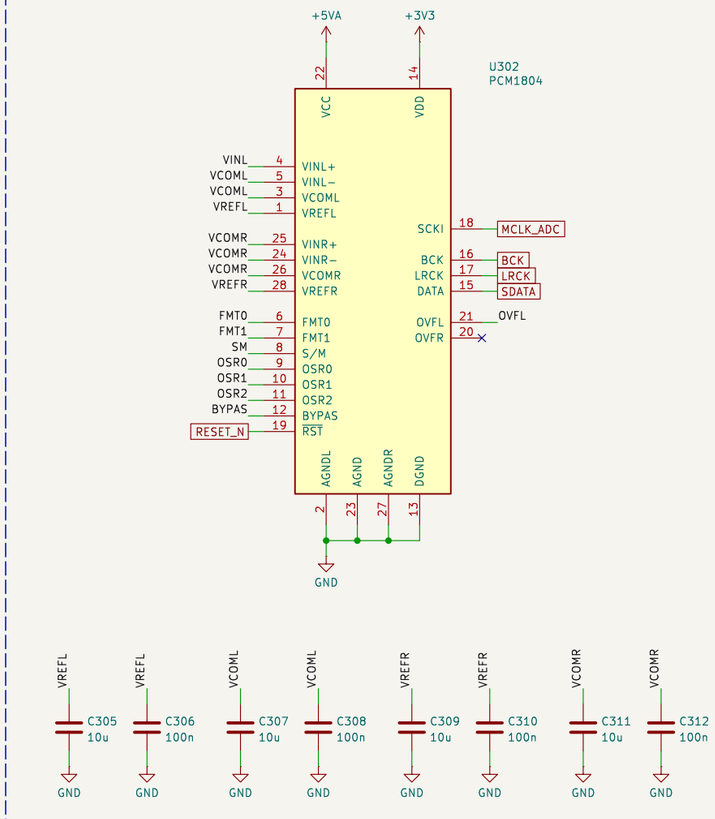


PCM1804 MODE (ON = 1)

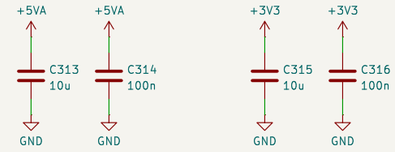


FMT0..OSR2: internal 51 k pull-downs.

PCM1804 master, 192 kHz



SUPPLY DECOUPLING

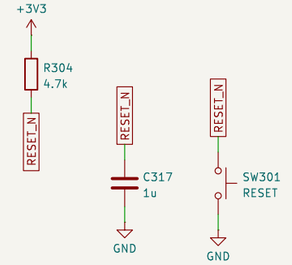


OVERLOAD FLAG



No LED (battery use):
meter or scope on TP301.

RESET



Default (192 kHz): S/M=0 master (switch 3 OFF), OSR2..0=111 (SCKI=128 fs), FMT1..0=01 I2S, BYPAS=1 (HPF off).
Change switches with power off, or press RESET afterwards.

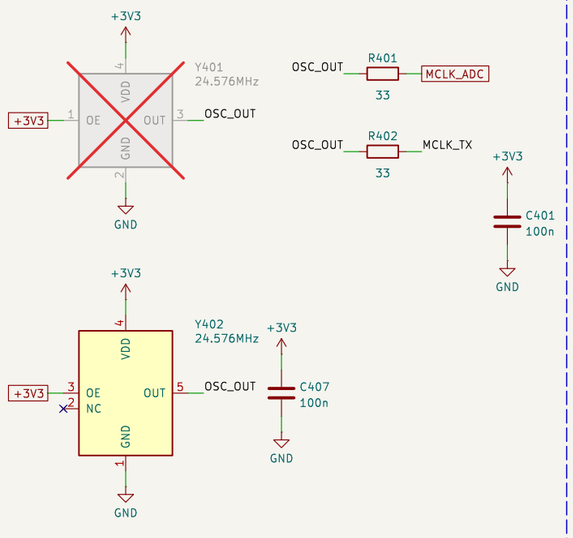
Generated by PCB/antenna_amplifier/design.py -- edit the script, not the sheets
CERN-OHL-W-2.0
ELARA -- ELF Atmospheric Radio Analyser

Sheet: /ADC (PCM1804)/
File: adc.kicad_sch

Title: ELARA antenna amplifier — ADC (PCM1804)

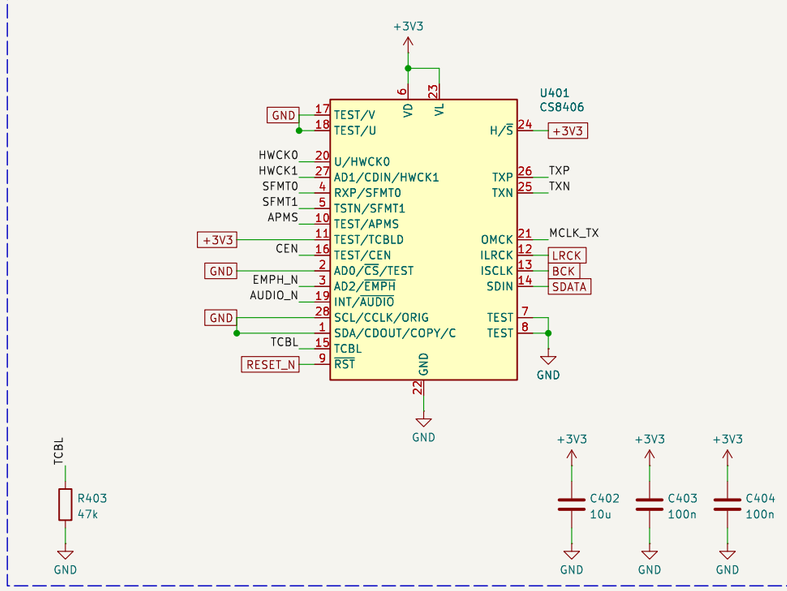
Size: A3	Date: 2026-09-30	**Rev: 0.2**
KiCad E.D.A. 9.0.9		Id: 3/5

MASTER CLOCK 24.576 MHz

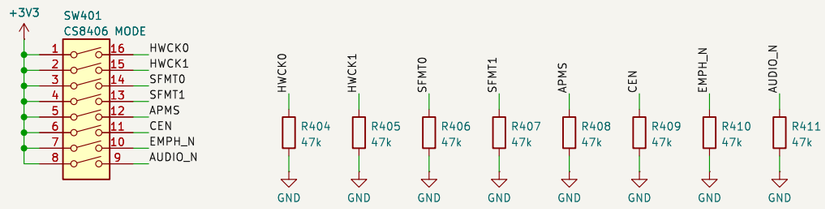


Fit Y402 (SOT23-5, leaded) or Y401 (3.2x2.5 no-lead), never both.

CS8406 hardware mode



CS8406 MODE (ON = 1)



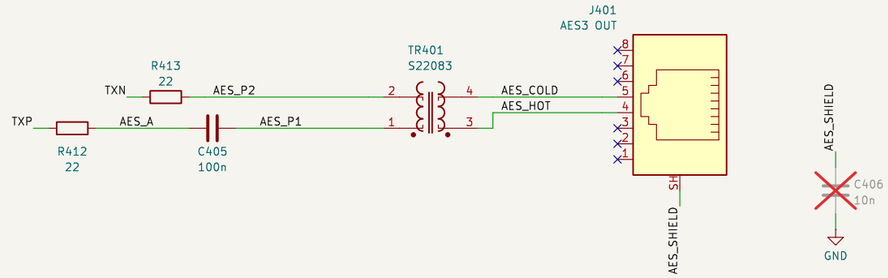
Default (192 kHz): HWCK -> OMCK = 128 fs, SFMT = I2S, APMS=0 (slave), CEN=0, EMPH_N=1, AUDIO_N=0.

TEST POINTS (clocks, reset, digital GND)



AES3 OUTPUT transformer isolated, RJ45 for shielded twisted pair

AES3 110 R balanced



AES3: 2 x 22 R + 2 x 33.5 R driver (CS8406 at VL = 3.3 V) = 111 R source, 3.3 Vpp into 110 R.

Cable: shielded Cat5e/6 (100 R pairs), shield connected at the indoor end only. AES3 is polarity-free.

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CERN-OHL-W-2.0

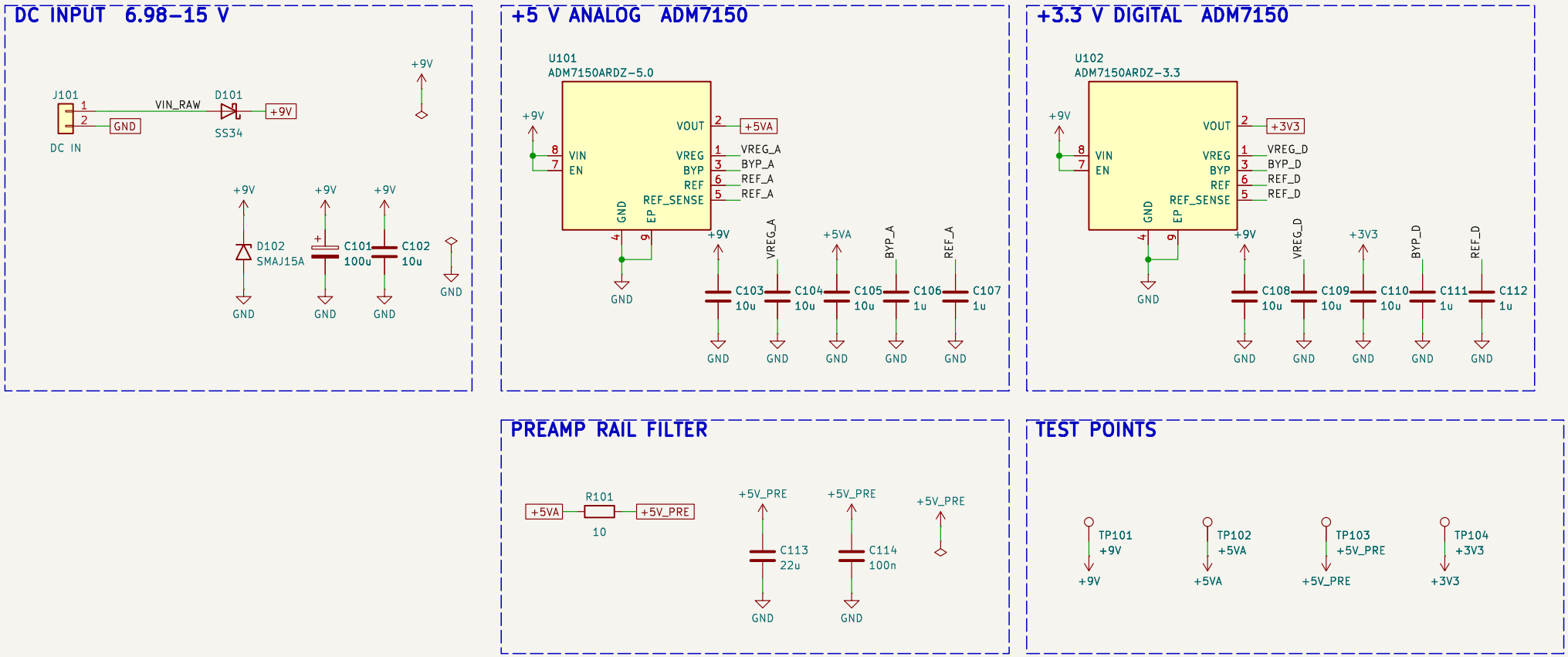
ELARA -- ELF Atmospheric Radio Analyser

Sheet: /Clock and AES3 output/

File: digital.kicad_sch

Title: ELARA antenna amplifier — Clock and AES3 output

Size: A3	Date: 2026-09-30	**Rev: 0.2**
KiCad E.D.A. 9.0.9		Id: 4/5



Budget: ~40 mA on +5VA (PCM1804 VCC, preamp, ADC driver), ~45 mA on +3V3.

ADM7150: REF_SENSE tied to REF (fixed output), EN tied to VIN.

No switching regulators on this board. +9V is the DC bus after D101 (about 6.0 V from the PSU, up to 15 V from a battery); the net keeps its rev 0.1 name.

No LEDs: the board may run from a battery. Check the rails at the test points with a voltmeter.

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CERN-OHL-W-2.0

ELARA -- ELF Atmospheric Radio Analyser

Sheet: /Power input and regulators/

File: power.kicad_sch

Title: ELARA antenna amplifier — Power input and regulators

Size: A4	Date: 2026-09-30	**Rev: 0.2**
KiCad E.D.A. 9.0.9		Id: 5/5